MOAIN AL-ABBASI

Software Engineer · Full-Stack Systems · AI Agents (LangGraph, MCP)

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SUMMARY

Software engineer building business systems and AI agents end to end in TypeScript, Python and Go, from database and API to web, mobile, hardware integration and deployment. Built a POS on an existing Oracle database with Bluetooth receipt printing, a ZKTeco attendance system with Saudi WPS payroll export, a multi-tenant notification platform, and Basira, an Arabic quotation verifier with 0/500 false alarms. Fully remote, Saudi time zone.

EXPERIENCE

Founder & Software Engineer · Alabbasi Tech (alabbasi.uk) · Sana'a, Yemen · Remote — 2026 – Present

- **POS replacing YemenSoft Onyx Pro** — web POS that works directly on the existing Oracle database: sales, returns, shifts and reports. Arabic receipts print on a Bixolon SPP-R310 over Bluetooth SPP through Web Serial; documented the failed Xprinter USB path on Windows. — NestJS 10, React 19, Oracle PL/SQL, BullMQ, OpenAPI
- **Motech Attendance** — replaces the ZKTeco desktop software: branch collectors pull punches from iFace950 devices over TCP into a central multi-tenant service with idempotent ingest; shifts, leave balances, CSV/Excel reports and WPS payroll export for Saudi Arabia (MHRSD/SAMA) and the UAE (SIF). Tested on a live device (3,118 punches). — Go, PostgreSQL, Python (pyzk), systemd
- **Nexus Notify** — multi-tenant, multi-channel notification platform behind one API, with an Angular dashboard and a native Android SMS gateway (dual SIM, foreground service). — NestJS, PostgreSQL 17, Prisma, Redis/BullMQ, Angular 22, Kotlin, Jetpack Compose
- **Secure remote access** — Zero-Trust platform with signed agents, per-user policies and session audit, plus a cross-platform single-command CLI agent that joins the NetBird mesh; releases signed with cosign in GitHub Actions. — Go, SSH, NetBird
- **Google Ads tooling for Keif Al-Diafa (Jeddah)** — persistent OAuth connector to Google Ads and Search Console with free-form GAQL queries and an Arabic dashboard; phase 1 of a multi-client ads platform with live Google Ads API data and a 12-table schema. — Hono, Cloudflare D1, Next.js 15, Prisma, PostgreSQL 16
- **keif-publisher** — generates posts from a media archive, routes each through a human approval gate, then publishes on schedule; dry-run and kill switch on by default. — Python, FastAPI, APScheduler
- Websites for Saudi hospitality brands: keifaldiafa.com, asoulaldiafa.com, qahwajeen.com. In development: an offline-first event-management PWA (Angular 22, IndexedDB) and a modern layer around a legacy PHP accounting system.

AI AGENTS & APPLIED AI

Basira · basirapp.site · github.com/MoTechSys/Project-Basira
Verifies Quran and Hadith quotations byte-exactly against 71,987 records from licensed corpora; an optional LLM only proposes spans and reads images (OCR), and never decides a verdict. 150/150 evaluation cases, 0/500 false alarms, 78.54% on IslamicEval 2025 subtask 1B (public dev set). 487 automated tests (345 backend, 110 bot, 32 frontend). REST API, MCP server and Telegram bot; entry to the AI in Service of Islamic Content Challenge 2026. — Python, FastAPI, React 19

Mansur AI (private repository)
Arabic multi-agent system: planner, ReAct executor, verifier and learner orchestrated with LangGraph; 13 agents, 38 tools, model router with fallback and cost tracking, hierarchical memory (working, episodic, procedural, semantic) and skills saved from successful runs. 82 passing tests. — Python, LangGraph, FastAPI, WebSocket, SQLite

Ai_Alabbasi
Coding agent with a model backend swapped from one config file (Claude, GPT-5, DeepSeek or a local model); task planner, CodeAct sandboxed Python, hybrid retrieval with FTS5 and vectors fused by RRF.

Moain Promoter AI
Seven specialised agents under an orchestrator that write bilingual Arabic/English posts for six platforms, with a Next.js dashboard. — Python, Celery, Redis, Playwright

MCP and agent tooling
Diafa Studio (Arabic PDF generation exposed to AI agents through MCP), a Telegram MCP server, and Bina, reusable agents for website audit, rebuild and Arabic SEO content.

SELECTED PROJECTS

Idhaat · github.com/MoTechSys/Idhaat-Platform
Live-tutoring platform: conflict-free allocation of four Zoom rooms with fallback to self-hosted Jitsi secured by a JWT for each lesson; recordings upload to Cloudflare R2 in chunks during class, play in a protected player with a moving watermark of the student's name, and are deleted after 48 hours. Runs for about 25 SAR a month. — Hono, Cloudflare Pages, D1, R2, Vitest

Grocery Ledger · github.com/MoTechSys/mohanad-web-app-2
Offline Android accounting and POS app for small shops: barcode by camera (ML Kit) or USB/Bluetooth scanner, append-only ledgers, PDF/Excel export. 157 passing tests, v2.2.1. — Flutter

Sijil · github.com/MoTechSys/Sijilati
Offline debt ledger for shop owners and workers, including those who cannot read: amounts entered as banknotes, spoken confirmation, 8-second undo. 93 passing tests. — Flutter

Ghaida · github.com/MoTechSys/ghaida
Household guidebook for domestic workers in their native language with AI voice for every sentence; offline PWA with admin panel.

S-ACM · github.com/MoTechSys/S-ACM-Project
Academic content-management system with AI summarisation and question generation. — React 19, TypeScript, Tailwind 4

GRADUATION PROJECT (TEAM)

Security Operations Center (SOC) · github.com/MoTechSys/my-bro
SOC built on open-source tools (Wazuh 4.14.7, Suricata, YARA, VirusTotal, auditd, MikroTik syslog) with an AI analyst whose actions require human approval. 14 use cases verified in a lab, including automatic blocking of SSH brute-force attacks. Digital forensics lab with Autopsy and FTK Imager.

TECHNICAL SKILLS

Programming	TypeScript, Python, Go, Dart, Kotlin, SQL, PL/SQL
Backend	NestJS, FastAPI, Hono, Prisma, REST/OpenAPI, WebSockets, Redis, BullMQ, Celery
Frontend & mobile	React 19, Next.js, Angular 22, Tailwind CSS, PWA, IndexedDB, Arabic RTL; Flutter, Jetpack Compose, ML Kit
Data	PostgreSQL, Oracle, SQLite, Cloudflare D1
AI & agents	LangGraph, multi-agent orchestration, MCP servers, RAG, hybrid search (FTS5, vectors, RRF), model routing; OpenAI, Anthropic, Gemini
Integration	ZKTeco (pyzk), ESC/POS via Bluetooth and Web Serial, Google Ads API
DevOps & security	Docker, GitHub Actions, Cloudflare, Linux; pytest, Vitest, Playwright; Wazuh, Suricata
Languages	Arabic (native), English (technical reading and writing)